BIKRANT KUMAR MISHRA

B.Sc. Geology | Aspiring Computational Geoscientist

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github.com/bikrantmishra2005

https://github.com/bikrantmishrageo2005/BIKRANTMISHRAGEO2005

EDUCATION

2023 - 2026 (Expected)

KHALLIKOTE UNITARY UNIVERSITY

- B.Sc. (Honours) in Geology
- **CGPA: 8.14/10**

2021 - 2023

HIGHER SECONDARY EDUCATION (SCIENCE)

- CHSE, Odisha
- **DISTRICT TOPPER**

2020 - 2021

SECONDARY EDUCATION (10TH)

- BSE, Odisha
- **RANK 5 AMONG CLASS**

LANGUAGES

- **English** — Professional Working Proficiency
- **Hindi** —Full Professional Proficiency
- **Odia (Odiya)** — Native Proficiency

PROFILE

I am a B.Sc. Geology undergraduate with a strong interest in Geo-AI, GIS, and computational geoscience. My academic work integrates geological field investigations with data analysis and machine learning techniques for applications in hazard assessment and resource exploration. I have worked on projects involving seismic data analysis, mineral prospectivity mapping, and environmental modeling. My long-term goal is to pursue advanced research in AI-assisted geoscience for sustainable resource management and disaster risk reduction.

RESEARCH INTRESTS

My research interest lies in AI- and GIS-integrated mineral prospectivity mapping, focusing on the computational exploration of critical and radioactive minerals. I aim to combine geospatial analysis, remote sensing, and machine learning techniques to develop predictive models that enhance data-driven mineral exploration and resource assessment.

SKILLS

1.Computational Geoscience & Geo-AI

- **AI-driven geological interpretation & predictive modeling**
- **Hydrocarbon & mineral prospectivity modeling (conceptual & ML-based)**

2.**Programming & Data Science**

- **Basic Python for geoscientific applications Scientific computing (NumPy, Pandas, Matplotlib) Supervised ML models (classification, anomaly detection, Random Forests)**

3.**Geospatial & Remote Sensing Analysis**

- **Basic GIS mapping & spatial data handling (QGIS- academic exposure)**

4.**Geological & Field Skills**

- **Geological mapping & structural analysis Lithology identification &**
- **weathering profile analysis Coastal & engineering geology observations**

5.**Modeling, Visualization & Scientific Communication**

- Basic 3**D geological and subsurface visualization (Blender)**
- **Data-driven visualization & technical reporting Research-**
- **oriented presentation & documentation**

6. Laboratory & Analytical Skills

- **Thin section preparation & petrographic microscope analysis (PPL/XPL)**
- **Optical mineral identification & texture analysis**
- **Basic geochemical data interpretation**
- **Sample preparation, documentation & lab safety protocols**

REPORTS

1.**College-Based | Submitted**

- **Geological Field Report on the Eastern Ghats Mobile Belt (EGMB), Odisha**

Field-based geological investigation focusing on lithology, structural features, metamorphic zones, and petrographic characteristics across Bhabanandha, Bhatakumuruda, Baghdevi, and Athagadapatana areas of Ganjam District, Odisha.

2.**SELF-INDEPENDENT | Submitted**

- **Coastal Geomorphology & Sediment Dynamics – Gopalpur Coast, Odisha**

Self-driven coastal geology report focusing on beach morphology, sediment characteristics, and shoreline processes.

- **Local Geological Assessment – Koraput (Jagannath Mandir Area), Odisha**

Independent field-based study of lithology, weathering characteristics, and stone material associated with heritage structures.

PROJECTS

1- AetherVision Omega – Geo-AI Smart City Intelligence System Self-developed Geo-AI framework integrating pollution monitoring, disaster early-warning, seismic risk analytics, and smart urban planning. Designed to model sustainable and resilient cities using geospatial data, machine-learning workflows, and 3D visualization concepts. (State-Level Hackathon – 3rd Prize)

2- **Earthquake Strength Classification Model (Machine Learning–Based) Developed a seismic event classification system to categorize earthquake strength and intensity using Python. Applied statistical analysis and machine-learning techniques to analyze seismic signals, magnitude distributions, and temporal patterns. (https://github.com/bikrantmishrageo2005/GeoAI-Earthquake-Strength-Classification)**

3- **Mineral Prospectivity Mapping Using Geo-AI**
Built a Geo-AI–based mineral prospectivity analysis model integrating geological parameters and machine-learning concepts to identify potential mineralized zones from spatial and lithological data.
(https://github.com/bikrantmishrageo2005/mineral-prospectivity-mapping)

4- **Hydrocarbon Prospectivity Modeling**
Developed a predictive model for hydrocarbon prospectivity using lithological, structural, and basin-related geological inputs.
(https://github.combikrantmishrageo2005/GeoAI-Hydrocarbon-Prospectivity-Modeling)

RESEARCH PAPERS & MANUSCRIPTS

Mishra, B.K. (2026). OMEGA-CITY: A Geo-AI–Powered Sustainable Urban Intelligence Framework for Real-Time Environmental Optimization, Earth-Hazard Prediction, and Climate Resilience.
Abstract provisionally accepted at the 1st International Conference on Sustainability-Oriented Learning for a Viable Earth (SOLVE 2026), VITAM, Odisha, India.

Work Experience

Academic Tutor – Geology
MMG Science Tutorial, Rayagada

Provided approximately six months of academic instruction in core geology subjects, demonstrating professionalism, subject competence, and strong student mentorship.